# MAPPING A RESEARCH AREA WITH AI AGENTS: 373 OPEN PROBLEMS IN ANALYSIS

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Abstract. In October 2026 we directed AI coding agents at every open problem we could find in one area of analysis: Hardy, Bergman and Dirichlet-series spaces, Hankel and Helson operators, and Nehari-type theorems. After a week of directed work on single problems, most of the campaign took two days. The agents collected 373 problems, found that 64 of them had already been settled by others, and produced claimed answers to 141 more. We classify these answers by how they were obtained and by how much they matter. Most are easy in hindsight: short arguments from known tools, results assembled from the literature, or counterexamples that are easy to check once found. Six are, in our judgement, of real significance. On the hardest problems, days of work by several agents produced maps of what fails rather than proofs. The work was directed by one person with a master's degree in the area who is not a research specialist. Our conclusion is that such a person can now map a research area, clear its long tail and find questions worth working on, but that it still takes an expert to judge which results matter, and a hard problem stays hard. A blind audit of a random sample of 20 answers found no mathematical errors, but two that answer less than was asked. All results of ours are sketches until they appear on arXiv or in a journal; this is a living document, updated as the people who asked the questions respond.

## 1. Introduction

Terence Tao has described research mathematics as climbing "a big mountain range with lots of tall mountains and lots of foothills" [53]. Human mathematicians climb one step at a time, but they can plan a route to a distant summit. In his picture, current AI tools are small jumping robots: they can sometimes "parkour their way to the top of a 6-foot wall" that a human could not climb, but they cannot plan the long route to a summit, and "the little jumping robots are nowhere near the Mount Everests of math" [53]. The robots are good at short, sharp obstacles, and the great problems remain out of reach.

This paper is a field report on what such robots do when pointed at an entire range. In October 2026 we used Claude Code and Codex agents to find, sort and attack every open problem we could locate in one research area. A week of directed work on single problems came first; the bulk of the campaign, over the whole database, took two days. The agents found 373 problems in papers, problem lists and monographs, established that 64 were already solved, and produced answers to 141 others.

The answers need careful reading. The agents were very good at mapping the field: finding problems, classifying them, checking the literature, and clearing problems that were easy but had received little attention. Most of their answers are low-hanging fruit; of the 141 we classify 6 as significant, 62 as useful and 73 as minor. On problems where experts had made partial progress and stopped, the agents produced reformulations, special cases and catalogues of obstructions rather than proofs. The gain was in cartography, not in climbing.

We did not solve 141 problems in the sense a mathematician would mean. The database mixes interesting questions from the field, questions at its research frontier, and questions we derived ourselves; many of the answers are to the last kind, or to questions that turned out to be easy once someone looked. Of almost 400 problems, about six of real significance were solved. What the agents added beyond that is partial: special cases and correct orders of growth for 142 problems, reductions of some 25 problems to three common hard cores, and

*Date*: October 14, 2026.

written records of why each attempted route failed. These are new insights into the problems rather than solutions of them.

The work was directed by one person with a master's degree in the area who is not a research specialist; most comparable reports come from teams of specialists (Section 12). Such a person can now explore a field at its frontier, but every result that mattered still needed an expert to recognise it, and for the most significant one the expert who asked the question became a coauthor. The askers' judgement remains the real test.

We therefore see this method as the first step of a research project: a fast way to get to know a field and to find questions worth working on. Section 3 describes how the problems were found and sorted, Section 4 the workflow, Section 5 the results, Section 7 a blind audit of a random sample, Section 8 the problems that were not settled, and Section 9 verification and credit. Sections 11 and 12 compare the work with other reports, and Section 15 lists its limits. The appendices give a sample record, how the record fields were used, what version control and file handling taught us, and the practices we adopted from other studies.

## 2. Setting

The area is the theory of Hardy and Bergman spaces on the disc and on the infinite polydisc, Dirichlet series viewed as functions on the infinite torus, and the Hankel, Helson and composition operators acting on them. A typical question asks for the sharp constant in an inequality, whether an extremal function can have zeros, or whether a bounded bilinear form has a bounded symbol. The active literature is a few hundred papers, much of it from a small number of research groups.

A typical question, and one of the six we discuss below, comes from Brevig and Saksman [4]. For $0 < p < 1$ the Hardy space $H^p$ consists of the analytic functions $f$ on the unit disc for which $\sup_{r<1} \int_0^{2\pi} |f(re^{it})|^p \, \mathrm{d}t/2\pi$ is finite; for $p < 1$ it is not a Banach space, and many standard tools fail. Among all $f$ in the unit ball of $H^p$, consider those that maximise the size of the $k$th Taylor coefficient. Brevig and Saksman conjectured that every such extremal function has no zeros in the disc. Zeros matter because a function without zeros can be written as a power $g^{2/p}$ of a function $g$ in the Hilbert space $H^2$, which turns the problem into a much better understood one. The conjecture is a precise, checkable statement, it had resisted the methods of its authors, and an answer would be noticed by the few dozen people who work on such problems. Most questions in our database are of this kind.

The author directed the work. He holds a master's degree in this area and is not a research specialist. He chose targets, judged whether a result mattered, steered agents that were stuck, read everything before it went out, and handled all contact with other mathematicians. The agents were Claude Code (Claude Opus 5.5 as the main session, with smaller Claude models for cheap subtasks) and Codex, with OpenAI models that Codex offered under the names GPT-6 Luna, GPT-6 Astra and GPT-6.1 Sol, often running at the same time in the same repository. The record of the work is the version history, with about 1000 commits, together with the logs: 14 Claude Code sessions with 393 subagents, and 586 Codex sessions, 420 of them subagents.

## 3. Building the map

The problems came from four sources, in this order.

(1) *Stated problems in a local paper collection.* Agents searched each paper for environments labelled Conjecture, Question, Problem or Open problem, and for phrases such as "remains open" and "it would be interesting to". Each candidate was recorded with the paper, the exact location, and a quote.
(2) *The papers those cite*, with the same extraction.
(3) *Synthesis passes.* Agents read the recent papers of the core authors "between the lines". This produced *implicit* questions, which a remark suggests without posing,

and *inferences*: paper A proves one statement and paper B another, so a third should follow.

(4) *Problem lists and monographs*: workshop problem sessions, a classical problem book, and the open-problem sections of two monographs.

**From candidates to 373 records.** Table 1 follows the problems from extraction to the database. The local collection was broad: the director's paper library also held books from unrelated projects, so the first catalogue contained questions from fluid dynamics, arithmetic geometry and combinatorics alongside the target area. Scope was then fixed by one sentence: *Hardy and Bergman spaces, Dirichlet series, analytic extremals, Hankel operators, sampling and interpolation, and directly related analytic number theory*, informally "what the people working in this area would work on". An agent applied it theme by theme and wrote down a reason for every excluded theme, for example "Euler and Navier–Stokes questions are outside the selected scope" or "sphere-packing and energy-optimality conjectures are excluded"; adjacent topics were kept only with a stated connection, such as random multiplicative functions for their link to Dirichlet-polynomial norms. Incomplete candidates, mostly exercises, were dropped. Excluded records were kept in a separate report rather than deleted.

| Stage (date) | Records |
|---|---|
| PDFs screened in the local collection | 299 |
| Candidate records extracted, deduplicated (10 Oct) | 459 |
| of which problem statements | 399 |
| of which incomplete candidates, dropped | 60 |
| Excluded as out of scope, with a reason per theme (11 Oct) | 166 |
| Kept in the focused catalogue, plus 2 local additions | 235 |
| first order (in the collection) / second order (cited by it) | 178 / 57 |
| Database at migration, adding 40 questions from our own book (12 Oct) | 270 |
| Added from synthesis, workshop lists and monographs (12 Oct) | 72 |
| Added during the work, mostly questions met while attacking others | 31 |
| Database | 373 |

TABLE 1. How the database was built. Duplicates were merged at each step, so the rows do not add up exactly.

Each problem became one Markdown file with validated metadata: its theme, its kind (298 stated, 55 implicit, 20 inferences) and origin (289 asked by others, 84 by us), tags, impact and difficulty scores, a dated literature check with quoted evidence, the stage reached in our own work, and, once anyone settled the problem, how far the resolution is trusted. Section A shows a complete record and which fields the agents filled at each stage. Moving from a single catalogue file to one file per problem let several agents and two different tools edit the database at once.

**How difficulty was scored.** Each problem received a difficulty score from 1 to 5, in two rounds. On 10 and 11 October, 194 problems from the first catalogue were assessed by cheap Codex agents (GPT-6 Luna), each with about ten minutes: a literature check, a short attempt, and a "speculative difficulty score and confidence" on the scale 1 = routine deduction or literature reconciliation, 2 = focused tractable project, 3 = substantial specialist project, 4 = probably requires a major new idea, 5 = longstanding frontier problem. Their brief warned that "a ten-minute failure to prove a statement is not evidence for a high score". On 12 October three Claude Opus 5.5 subagents scored the whole database, each taking a group of themes, under a common brief that defined difficulty as "effort for us with current tools" and fixed an anchor problem at each level:

| Difficulty | Meaning |
|---|---|
| 1 | short argument from known tools; days |
| 2 | focused effort; an existing method adapts, or a computation settles it |
| 3 | needs a genuinely new idea, but a route is visible |
| 4 | hard; the natural approaches are known to fail |
| 5 | major open problem, or equivalent to one |

Each score came with a one-line justification. The agents read each record's statement, literature status and evidence, and the source where the statement was unclear; resolved problems were scored as originally posed. For the 189 problems that already had a score from the first round, the Opus agents kept it unchanged; they scored the rest from scratch. Over all 373 problems the initial scores were distributed as 37, 48, 110, 145 and 33 for difficulty 1 to 5. The scores moved during the work. After the scoring pass, 145 of the 168 problems that remained unsettled were re-rated, almost always upwards (127 by one level and 23 by two), as attempts failed. Most of these moves were mechanical: the protocol raised any problem that used its session budget without a full answer to difficulty 5 (see below). Of the 141 problems our agents settled, 38 were re-rated, mostly downwards. The final scores therefore record the outcome rather than predict it, and we use the initial scores of 12 October throughout. Even these are not blind: they were assigned after the directed work of 3 to 11 October, so for problems attempted then the score already reflects that attempt.

**Selection for attack.** Problems were attacked in rounds by difficulty, from 1 upwards: "start with the low-hanging fruit". Within a round the orchestrating session picked open, unparked problems with the best impact relative to difficulty, mixing cheap problems with one or two of high impact, and skipping problems whose literature status was an unverified claim by others. Each problem had a budget of working sessions (one agent attempt, typically up to an hour) equal to its difficulty, from one session at difficulty 1 to four at difficulty 4; difficulty 5 problems were worked only in dedicated rounds with a specific new idea. The protocol then moved problems mechanically: a problem at difficulty 1 to 4 that used its budget without a full answer was raised to difficulty 5, and a problem was parked once it reached difficulty 5, or once it had used six sessions without even a partial result. Parked problems, which are removed from the queue, could be reopened only for a new idea, paper or tool.

The public problem volume was a further selection: questions asked only internally were removed, leaving 308 questions and 58 answers.

## 4. The workflow

Each problem was worked in sessions. A session begins by reading the record, the source passage and the log of earlier attempts, and by writing a three-line plan. It ends by rewriting a short *state of play* and appending one line to the *attempt log*: the route, the outcome, the reason, and the next step. A claimed result had to become a *solution sheet*: the problem as posed, the answer as a theorem, a complete proof with exact citations, and nothing else, built through a LaTeX pipeline that refuses to compile files that fail an automatic style check. A fresh agent with a fixed brief then refereed it. The referee re-read the question in its source, re-derived each step, reran any certificate, and decided whether the answer was full or partial.

On hard problems several agents worked in parallel, each assigned one approach (we called these *lanes*), while a coordinating session read their interim reports and redirected them. When the agents drifted into cataloguing obstructions, a stricter brief (*prove mode*) told each to commit to one strategy and push it to a theorem, or to a proof complete except for one precisely stated lemma. The full instructions are published with this paper as a reusable set of agent playbooks.

How each record field was used during the work, and which ones mattered, is analysed in Section B.

## 5. Results

5.1. **What the literature check found.** The initial difficulty score predicted the outcome (Table 2): over all 373 problems, a lower score went with being settled, with an area under the ROC curve of 0.79 (FAR reports 0.69 for its difficulty scores [14]). The figure flatters the scores. They were not blind, since many problems had been tried before they were scored, and problems were attacked in order of difficulty, so the easy ones also received attention first. The table also shows why easy problems are easy: of the 37 problems rated easiest, 15 already had a published answer and our agents settled the other 22. Tao observed the same correlation on the Erdős problems: the problems current tools can solve are "precisely the category of nominally open problems that are most likely to have been solved in the literature" [58]. The literature check also caught one problem recorded as resolved by a paper that had since been withdrawn.

| Initial difficulty | Settled by others | Settled by us | Partial | Open | Total |
|---|---|---|---|---|---|
| 1 | 15 | 22 | 0 | 0 | 37 |
| 2 | 12 | 34 | 2 | 0 | 48 |
| 3 | 20 | 50 | 32 | 8 | 110 |
| 4 | 10 | 32 | 91 | 12 | 145 |
| 5 | 7 | 3 | 17 | 6 | 33 |
| Total | 64 | 141 | 142 | 26 | 373 |

Table 2. Outcome of all 373 problems by the first difficulty score each received (Section 3). "Settled by us" counts claimed answers that passed one agent referee.

5.2. **When the results came.** Figure 1 shows when the 141 answers were obtained, coloured by the initial difficulty. The database itself existed only from 12 October, so everything obtained in the directed work of 3 to 11 October appears in the first bar. That bar holds 30 answers, including four of the six we rate most significant. The bulk campaign of 12 October settled 45 problems, almost all of difficulty 1 to 3 and none of high significance. On 13 October the agents worked through harder problems: of its 66 answers, 25 were rated 4 or 5, and two are of high significance.

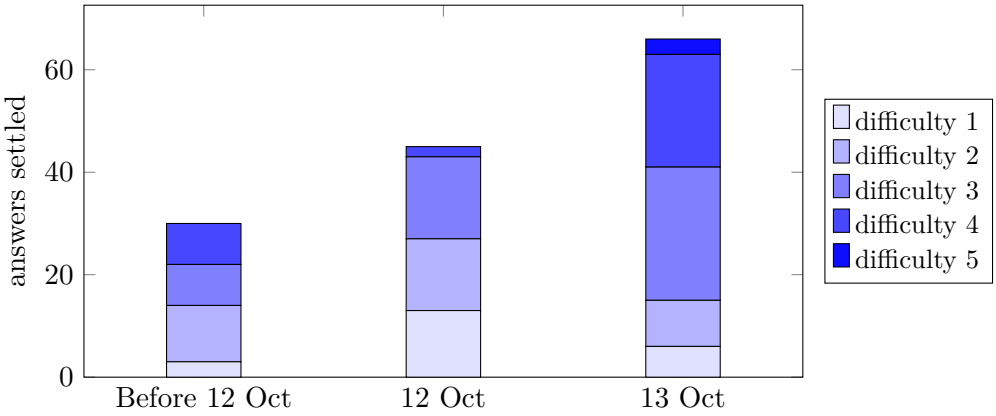



Figure 1. Answers settled by our agents, by when they were recorded and by the initial difficulty score. The first bar collects the directed work before the database existed; its dates come from the first appearance of each answer's write-up in the version history.

5.3. **What kind of results were these?** We classified each of the 141 answers along three axes.

- *Sign*: 82 proofs, 42 counterexamples, and 17 determinations of a value, set or order.
- *Kind*, meaning how the answer was obtained:
  - 20 assembled from the literature;
  - 58 routine new arguments;
  - 44 explicit constructions or certificates, which are the work to find and easy to check;
  - 19 substantial new arguments.
- *Significance*, judged by the mathematics alone:
  - 6 high: an answer to a question of standing in the field, at the level of a paper;
  - 62 medium: a full answer worth a short note;
  - 73 low: a corollary, an elementary or conditional answer, or an answer to a question of little interest.

The author chose the six high-significance results. The split between medium and low is a preliminary classification made by an agent from the records and checked only by spot, so we use it only for broad comparisons. Tables 3 and 4 give the cross-tabulations.

| | Literature | Routine | Construction | Substantial | Total |
|---|---|---|---|---|---|
| Proof | 17 | 43 | 5 | 17 | 82 |
| Counterexample | 0 | 1 | 39 | 2 | 42 |
| Determination | 3 | 14 | 0 | 0 | 17 |

TABLE 3. The 141 claimed answers by sign and by how they were obtained.

| | High | Medium | Low | Total |
|---|---|---|---|---|
| Proof | 4 | 33 | 45 | 82 |
| Counterexample | 2 | 24 | 16 | 42 |
| Determination | 0 | 5 | 12 | 17 |
| Asked by others | 6 | 57 | 26 | 89 |
| Asked by us | 0 | 5 | 47 | 52 |

TABLE 4. The 141 claimed answers by significance, split by sign (top) and by who asked the question (bottom).

Three things stand out. Counterexamples are where the agents are strongest: 39 of the 42 counterexamples are explicit constructions or certificates. Searching a space of candidates and checking each one is exactly what agents do well, and what earlier work on machine-assisted discovery exploited [33, 45, 60]. Few proofs are substantial: 17 of the 82 needed a genuinely new argument, 43 are routine, 17 are assembled from the literature and 5 rest on a construction. And the significant results came from directed work. Four of the six were obtained in the week of directed work before the database existed, while the author steered one problem at a time with the asker's question in view; the other two came from the campaign over the database.

5.4. **The six significant results.** Table 5 lists the six results with their sources and their current status.

Stated precisely:

(1) For $0 < p < 1$, every extremal function of a nonzero bounded linear functional on $H^p$ is outer; in particular the extremals for the coefficient problem of Brevig and

| Result | Question | Checked by | Askers told |
|---|---|---|---|
| Extremals on $H^p$, $0 < p < 1$, are outer | See [4, Conj. 1] | agent, Lean, coauthor | coauthor |
| At most one zero | See [3, Question 1] | agents, Lean | yes |
| Bernstein–Nikolskii limits | See [6, Conj. 1.10] | agent, not filed | no |
| $H^1$ partial sums | See [54, Problem 4.5] | agent | no |
| Positive Nehari fails | See [7, after Rem. 4.11] | agents | yes |
| Approximant zeros | See [5, Conj. 7.1] | agent | no |

TABLE 5. The six significant results. "Askers told" means the result was sent to an asker or, in one batched note, to a colleague in the askers' department.

Saksman have no zeros in the disc, which answers their conjecture in a stronger form. This was written jointly with Brevig.

(2) For $0 < p < 1$, $k \geq 1$ and $0 \leq t \leq 1$, every extremal for the prescribed-coefficient problem of Brevig, Grepstad and Instanes has at most one zero in the disc, counting multiplicity.

(3) For $0 < p < q < \infty$ and every $s \geq 0$, the sharp Bernstein–Nikolskii constants for the $s$th derivative of polynomials of degree $n$, suitably normalised, converge as $n \to \infty$ to the corresponding constant for entire functions of exponential type, for both complex and real polynomials (Ganzburg and Tikhonov's conjecture).

(4) The norm of the partial-sum operator $S_N$ on the Hardy space $\mathcal{H}^1$ of Dirichlet series is of order $\log N / \log\log N$.

(5) There is a bounded nonnegative Helson matrix without a bounded symbol, so the positive-symbol version of Nehari's theorem asked about by Perfekt and Pushnitski fails.

(6) For every $1 < p < \infty$ with $p \neq 2$ there is an outer polynomial target whose degree-one optimal polynomial approximant in $H^p$ has a zero inside the disc, contrary to Centner's conjecture.

None of the six is yet on arXiv.

5.5. **How the question was posed.** The database mixes questions of very different standing: conjectures, where an author predicts the answer; numbered problems and questions, including those from problem sessions and monographs; informal remarks such as "it would be interesting to know"; and questions we derived ourselves. The records did not store this distinction, so we recovered it afterwards from each record's source location; the split is approximate.

| How posed | Records | Others solved | We solved | Unsettled | High |
|---|---|---|---|---|---|
| Conjecture (others) | 63 | 13 | 23 | 27 | 3 |
| Problem (others) | 69 | 18 | 18 | 33 | 1 |
| Question (others) | 67 | 12 | 24 | 31 | 2 |
| Informal remark (others) | 90 | 18 | 24 | 48 | 0 |
| Our own question | 84 | 3 | 52 | 29 | 0 |

Our own questions were settled most often, 52 of 84, and none of those answers is significant. All six significant results answer a conjecture, problem or question that someone else had posed formally. Informally posed questions were the least often settled and produced nothing of high significance. We now record the form of each question when it is harvested.

## 6. Six discoveries, in the agents' words

The session logs record what each agent wrote when it reported a result. We quote them verbatim, with model and time (UTC); trims are marked [...] and our insertions are in square brackets. Table 6 gives the order.

| Result | First reported | Found by | Sign |
|---|---|---|---|
| Outer extremals | 8 Oct, 13:56 | Claude Opus 5.5 | proof |
| At most one zero | 8 Oct, 20:49 | GPT-6 Luna (Codex) | proof |
| Approximant zeros | 10 Oct, 21:46 | GPT-6.1 Sol; Claude | counterex. |
| Bernstein–Nikolskii limits | 11 Oct, 19:13 | Claude Opus 5.5 | proof |
| $H^1$ partial sums | 12 Oct, 22:21 | GPT-6.1 Sol (Codex) | proof |
| Positive Nehari | 12 Oct, 22:44 | GPT-6 Astra | counterex. |

TABLE 6. When the six high-significance results were first reported. Each was later checked by at least one other agent; the first two are also Lean-verified.

**Outer extremals.** The first claim came after the director told the agent that its numerical sweeps did not get closer to a proof:

> I believe I've proved [the nonvanishing conjecture of Brevig and Saksman]: for $0 < p < 1$, every extremal for $C(k, p)$ has no zeros in the disc. [...] Only I have checked the proof, plus numerics at $k = 2$; no one else has. —Claude Opus 5.5, 8 October, 13:56

Two hours later a subagent extended it from one coefficient to every bounded linear functional, including singular inner factors. The next evening a Codex agent estimated that a Lean formalization would take "months, potentially longer"; it was finished overnight:

> Done—the full theorem is now Lean-verified: every extremal for a nonzero bounded complex-linear functional on $H^p$, for every real $0 < p < 1$, is outer. —GPT-6.1 Sol, 10 October, 00:54

**At most one zero.** A Codex subagent found the key step; the deciding idea was to fix one perturbation phase rather than average over phases:

> The full real-direction Hessian yields a direct contradiction for any extremal with at least two zeros. [...] I checked the projections, the cancellation of the [...] terms, and admissibility of the perturbation. I found no gap in the argument. —GPT-6 Luna, 8 October, 20:49

The idea was relayed to the Claude session within minutes, which then reported that it had "reached the same conclusion independently". We cannot verify that claim, and we credit the Codex subagent.

**Zeros of optimal polynomial approximants.** This story took one evening. At 21:58 the director asked how close the agents were to the full result:

> We're not close to the full result if that means interior zeros for every $p \neq 2$. What we have is solid for $p$ close to 1, and nothing yet for $p$ near 2 or above it. —Claude Opus 5.5, 10 October, 21:58

Twenty-seven minutes later a subagent reported "there is no obstruction", and the main session wrote:

> Outer targets now work for $p > 2$ as well. Agent A found the construction, and I checked it independently. So for every $p \neq 2$ there is an outer target (even a polynomial one) whose degree-one OPA has a zero in $\mathbb{D}$. —Claude Opus 5.5, 10 October, 22:25

**Bernstein–Nikolskii limits.**

> [The finite-$q$ problem] is proved: for every $s \geq 0$ and $0 < p < q < \infty$, $\lim A^{(s)}_{p,q,n} = D^{(s)}_{p,q}$, and the real version holds too. I checked the proof step by step; it has not been refereed. —Claude Opus 5.5, 11 October, 19:13

A second subagent then checked priority and found the problem still listed as open in a survey from two weeks earlier.

**$H^1$ partial sums.**

> The draft now proves a lower bound of order $\log N/\log\log N$ for every sufficiently large $N$, matching the published upper bound. The key step is turning product entropy into the missing $L^1$ output bound [...] subject to independent review. —GPT-6.1 Sol, 12 October, 22:21

A Claude referee later judged it "a genuine full-order answer to the $p = 1$ case of [an open problem of] Saksman–Seip" [54, Problem 4.5].

**Positive Nehari.** The same agent declined to count a numerical candidate before it had a certificate, and four minutes later had one:

> I'm not counting the new [...] numerical candidate as a breakthrough—it still needs a rigorous certificate. —GPT-6 Astra, 12 October, 22:40

> [The record] now has a rigorous counterexample certificate. For an explicit measure on two points in the real three-dimensional cube, the certified scalar norm exceeds the full Hilbert embedding norm. [...] This is a proved candidate awaiting independent review, rather than an 'almost proved' lead. —GPT-6 Astra, 12 October, 22:44

A Claude referee reran the certificate the next morning and found it "correct, no fixes".

**The agents on their own work.** The same logs show the agents assessing their limits frankly once challenged:

> I did a bounded first pass, not an all-out attempt. [...] I tested one route per open problem, often a relatively elementary construction, and usually stopped when it failed. [...] 14 failed routes do not demonstrate that I seriously exhausted the mathematical possibilities. —GPT-6 Astra, 13 October

> Against the original questions: 0 of these 4 fully solved. I made substantive partial progress, but selected easier subproblems instead of problems I could plausibly finish. —GPT-6 Astra, 13 October

> When everything you try collapses onto the same estimate, it usually means the problem needs a new idea, not more effort. —Claude Opus 5.5, 13 October

## 7. A blind audit of a random sample

To estimate how far the counts can be trusted, we drew a random sample of 20 of the 141 claimed answers and 10 of the 64 problems marked as settled by others. Five fresh referee agents (Claude Opus 5.5) each took a group. They saw only the question as posed and the solution sheet or the cited paper, never the working notes or earlier verdicts, and spent 15 to 20 minutes on each answer.

Of the 20 answers, 18 were judged correct and complete, 2 correct but answering less than was asked, and none was flawed. With 20 answers this bounds the rate of flawed answers by about 17% at 95% confidence. The two downgrades are typical of the scope problem we saw throughout. One answered one part of a two-part question, and the other is unconditional only for part of the parameter range, relying for the rest on an unrefereed preprint, as it discloses. Four of the 18 correct answers depend on unrefereed preprints and say so.

The audit found more problems in the bookkeeping than in the mathematics. In five of the 20 records the metadata did not match the sheet: a dependency on another of our sheets was not listed, the listed write-up covered only a special case while the general proof sat in another file, two records left an unrefereed dependency out of the field meant to record it, and one record posed no explicit question. Of the 10 problems marked as settled by others, 8

were confirmed. Two are only partly settled by the cited work. In one the cited result covers part of the range and rests on a single-author preprint claiming a major open isoperimetric conjecture. In the other the problem is settled, but by papers the record does not cite.

These numbers are encouraging but limited. The referees come from the same model family that produced many of the answers, 15 to 20 minutes is enough to re-derive a short argument but not to referee a long one, and the sample is small. The audit supports the claim that the arguments the workflow produces are usually sound; it does not replace review by people in the field.

## 8. What the agents did not settle

Full answers are only half the picture. Of the 373 problems, 168 remain unsettled: 142 with partial results of ours and 26 without. We compared them with the 141 settled ones; the counts below come from the records, and the classification of obstructions is our judgement after reading each record's state of play.

### 8.1. **How they differ.**

| | Unsettled (168) | Settled (141) |
|---|---|---|
| Initial difficulty 4 or 5 | 126 | 35 |
| Asked by others | 83% | 63% |
| Impact 4 or more | 19% | 4% |
| Partial progress by others already in the literature | 50% | 21% |
| Answer is a counterexample (or expected to be one) | 5% | 30% |
| Inferences of ours | 1 | 16 |
| Mean number of working sessions | 1.8 | 1.5 |

Most unsettled problems received little work: 67 had one working session, 74 had two, and only ten had four or more. The settled problems received about as much, so effort alone does not separate the two groups; they differ in kind. They were asked by others, they matter more, half of them had already been attacked in the literature, and they ask for a characterisation, an exact constant or a uniform estimate rather than a single construction. The settled problems, by contrast, include most of our own derived questions and most of the problems a counterexample could close. Themes rooted in number theory and probability (random multiplicative functions, Bohr radii, general Dirichlet series, Paley–Wiener extremals) were mostly unsettled; composition operators and the extremal problems around one published conjecture were mostly settled.

### 8.2. **What blocked them.**

| Main obstruction | Records |
|---|---|
| Only special cases or families proved | 59 |
| Needs an estimate that is itself a known hard problem | 37 |
| Open-ended, ambiguous, or in need of reformulation | 23 |
| Endpoint or critical case; every method loses a log or a power | 22 |
| Exact constant wanted; only bounds or expansions found | 14 |
| Numerics suggest the answer, but no route to a proof | 6 |
| Depends on unrefereed or unavailable literature | 5 |
| Not attempted in depth | 2 |

About 25 records reduce to three underlying problems: an $L^1$ endpoint estimate behind Schatten-class Nehari theorems, the value of one sharp constant in a multiplicative Hilbert-type inequality, and the unknown extremal functions of point evaluation in Paley–Wiener spaces. Settling any one of these would move many records at once; none moved.

8.3. **Full versus partial.** A full answer settles the question as posed; a partial one does not. Of the 142 partial results, the main type was:

| Type of partial result | Records |
|---|---|
| Special case, subrange or family | 90 |
| Counterexample to a stronger statement or a literal reading | 15 |
| Correct order, or two-sided bounds, without the constant | 13 |
| Reduction to an equivalent reformulation | 12 |
| Conditional on a hypothesis or an unrefereed preprint | 7 |
| One direction only | 5 |

A recurring kind is a local result near a point where the problem is already solved, for example near $p = 2$, which proves the conjecture in a neighbourhood but gives no uniform control away from it. This is the pattern the human director called "filling $\mathbb{R}$ with $\mathbb{Q}$": more and more special cases, none of which converges to the general statement.

8.4. **Walls.** Only two problems received sustained work over several days. The first is the endpoint $p = 4$ of Nehari's theorem in the Schatten class $S_p$ for Hankel operators on the Paley–Wiener space of the disc, which Bampouras singled out after proving the theorem for $p > 4$ [1]. Over three days, up to seven agents worked on it in parallel, each assigned one approach. Every route reduced to one square-function estimate in $L^1$ that would be a theorem in its own right. The agents proved many special cases and wrote down equivalent reformulations, but no proof, and the problem was handed back to Bampouras with a note describing these reductions. The second is Ramberg's conjecture [2, 8] that restricting a function on the bidisc to the diagonal is a contraction from the Hardy space $H^p$ of the bidisc to the Bergman space $A^p$ of the disc for every $p \geq 2$; it is known for even $p$. In four sessions over three days the agents proved it for the functions whose diagonal is $z$ (and their Möbius images), for the diagonals $z^m$ when $p \geq 4$, and refuted a natural strengthening in terms of convex order; the general case did not move. In both cases the output is a useful map of the problem. It matches Tao's image: some walls of six feet were climbed, and the Everests were not approached.

## 9. Verification and credit

Producing a candidate proof was cheap; checking it was the bottleneck. This matches Tao's analysis of the problem-solving pipeline, in which generation is followed by verification, exposition, acceptance and absorption into the standard theory, and in which AI threatens "proof indigestion" [55]. Our answers were checked by agent referees, in some cases by a second model family, and in a few cases by Lean formalization or exact-arithmetic certificates. Agent referees caught real errors: a bound used outside its range, a missing hypothesis, and above all overstated scope, where an answer to a special case was reported as an answer to the question. They rarely asked whether the claim was the right one.

For this reason every result of ours is labelled a *sketch* until it appears on arXiv or in a journal, however it was checked. Before anything is posted, the people who asked the question are told; this follows the recommendations of the Leiden Declaration on attribution, disclosure and the humanity of authorship [50]. The questions belong to those who asked them. The proofs and computations were produced by AI agents and checked by the author. Following Tao's rule of thumb [55], no result should be published until a human can give a clear, expert-level talk on it.

## 10. History and files

*Version control was the research log, but only once we used it.* The repository was managed with Jujutsu (`jj`), which recorded 2409 operations and 1007 commits. Of the operations, 1944 fall on 12 and 13 October, after the rule that each problem record is committed together

with the work it describes. The directed work of 3 to 11 October, which produced four of the six significant results, left only 136 operations; those results can be dated only by when their write-ups first appeared. Everything quantitative in this report (when problems were settled, how difficulty scores moved, which record fields were used) comes from the fine-grained part of the history. Jujutsu's automatic snapshots meant that no edit by an agent was ever lost, and cheap small commits kept changes per record. Our playbooks therefore default to Jujutsu with a git backend, so that git-based hosting keeps working, and require a commit per problem from the first day.

*Failed attempts sprawled.* The rule that TeX files hold results only, with failures recorded in one line of the problem record, was adopted but not enforced. The research folders ended with about 350 Markdown notes, 575 TeX files, 260 scripts and 140 data files. 107 problem records list five or more artifacts, and one hard problem accumulated 31 files. Parallel agents each wrote their own note instead of appending to a shared one, superseded drafts were archived rather than deleted, and exploratory scripts were saved next to certificates. Later agents had to read through the pile to avoid repeating routes, and sometimes did not. The playbooks now require one notes file per problem with dated sections, scratch work outside the repository, scripts in the repository only when a claim depends on them, and deletion of superseded drafts, which the history keeps.

## 11. Related work

**Agents on lists of open problems.** Several groups have run AI systems over whole lists of open problems in the past year, and their numbers are the natural comparison for ours (Table 7). The closest study to ours is the Gemini case study on the Erdős problems, which

| Study | Problems | Outcome |
|---|---|---|
| Gemini on Erdős problems [15] | 700 open | 13 addressed, 4 novel |
| Formal proof search [25] | 353 Erdős; 492 OEIS | 9; 44 resolved |
| OEIS Open [9] | 492 OEIS | 147 resolved |
| Provers and verifiers [21] | 66 conjectures | about a third |
| QED [23] | 18 expert-chosen | 5 original works |
| FAR pipeline [14] | 4717 harvested | 598 candidate resolutions |
| UQ [26] | 500 unsolved questions | 15% pass validation |
| This report | 373 harvested | 64 known; 141 claimed; 6 significant |

Table 7. Reported outcomes of AI systems run on lists of open problems. The lists differ widely in difficulty, and so do the meanings of "resolved".

reached the same conclusion: "the 'Open' status of the problems was through obscurity rather than difficulty" [15]. Of the 13 problems it addressed, 9 turned out to be solved already in the literature, as were nine of our twelve easiest problems; of 200 candidate answers it graded, 68.5% were flawed and only 6.5% meaningfully correct, because most technically correct answers had misread the question and were vacuous. The FAR pipeline found that its difficulty scores predicted outcomes with an area under the curve of 0.69, a weak but real signal like the one in Table 2. The authors of the OEIS study note that their conjectures "are of uncertain mathematical significance, and most have likely received little previous attention" [9]. The FAR pipeline [14] is the nearest match to our method, at about fifteen times our scale: it extracted 6453 candidate problems from 5245 combinatorics papers and sent 77 potential resolutions to their authors. Like us, its authors identify the selection of problems and the review of the results as the bottlenecks. An early small test with five easy conjectures found the same pattern as ours: the easy ones fell, and the one that needed two papers combined did not [17]. Individual resolutions from such sweeps have been written up with care [11, 24], and one Erdős prize problem thought open for half a century turned out to have been settled three decades before it was posed [10].

**Harvesting problems.** Problem lists are now being built automatically. DeepMind's formal conjectures collect 2615 statements in Lean, including 1029 open conjectures; AI proofs and disproofs were used to find badly stated problems [16]. OpenConjecture extracts conjectures from new arXiv papers with a language model and ranks them by apparent tractability, with the caveat that the results "have not been independently verified" [32]. FrontierMath's open-problems pilot sets out to provide a "denominator" for AI contributions [30], and MathArena tracks performance on questions extracted from recent papers [22]. The Erdős problems database [40] and its AI wiki [41] remain the closest analogue to our database; the wiki warns that "absence of past progress may reflect obscurity rather than difficulty".

**Mathematicians' experience.** Gowers writes that "if a problem can be solved by means of a relatively standard argument, it is highly likely that an LLM will be able to find and use that argument", while its suggestions on harder problems often "sound promising until I think about them carefully" [31]. That is our experience in one sentence. Case studies from industry [13, 28] and from individual groups, including one in functional analysis [12] and one with a consumer model [27], report the same division of labour: models search for proofs, and experts close them. Earlier work on machine-assisted discovery found constructions and counterexamples by search [33, 39, 45, 46, 60], and experiments with frontier models as research assistants are collected in [37]. Formal proof [34, 35, 49, 51, 52] and evaluation [38, 42–44, 48] form the background.

**Verification and significance.** Bloom, who maintains the Erdős problems database, reports "a wave of AI-produced solutions provided with no explanation", and that much recent progress came from renewed human effort "rather than just AI" [29]. Kallel and El Louadi separate the validity of a derivation, which a proof checker settles, from whether the formal statement means the intended question and whether the result matters; "only the first is mechanizable" [19]. This is our distinction between settled and significant. A survey of arXiv mathematics finds substantive AI use growing to about 14% of submissions by August 2026, concentrated in combinatorics rather than analysis [18], and long agent runs are documented to drift in the ways we saw [20]. On the goals of mathematics beyond problem solving we follow Thurston [59], Gowers [47], Tao [55, 57] and Bessis [36]; journalism on the current state includes [53], where Igor Pak notes that language models are "fantastic in finding the right references".

## 12. How the method differs

Table 8 compares the design of our campaign with the closest studies. Three differences matter for reading our numbers.

| Study | Problem source | Who directed | Checked by |
|---|---|---|---|
| Gemini, Erdős [15] | curated list | expert team | AI filter, experts |
| Proof search [25] | curated lists | research team | Lean |
| OEIS Open [9] | formalized list | one researcher | Lean |
| FAR [14] | 5245 papers | research team | sent to authors |
| QED [23] | chosen by experts | experts | verifier agent |
| This report | harvested, about 300 papers | one MSc holder | agent referees; askers |

Table 8. How comparable studies chose their problems, who directed the work, and how results were checked.

*The problems were harvested, not curated.* The Erdős and OEIS lists are fixed, public and of known provenance. Ours was assembled by agents from papers, problem sessions and monographs, including informal remarks and our own derived questions, and it was scoped by a single sentence. This gives a broader and messier denominator; Section 5 shows how much the outcome depends on how a question was posed.

*The area resists formalization.* Most of the large studies verify in Lean, which works well for combinatorics and number theory, where substantive AI use is also concentrated [18]. Hardy spaces, Schatten classes and Dirichlet series in infinitely many variables are far from current formal libraries. We formalized two results; everything else rests on agent referees, certificates in exact arithmetic, and the people who asked the questions. Our counts are therefore claims, not verified resolutions.

*One non-expert director.* The other studies are run by teams of specialists or by the developers of the models. Ours was directed by one person with a master's degree in the area, which is closer to how a graduate student or a mathematician from a neighbouring field would use these tools. Three of the six significant results came from sessions where he directed a single problem with its asker in view.

## 13. What else the work produced

The proofs are the smallest part of what the campaign left behind. For a mathematician entering one of these problems, the other products are probably worth more. The first is a map of the area: 373 problems with exact sources, 308 of them published with their sources and evidence status. The second is a literature audit, which found 64 problems already solved and one recorded resolution that rested on a withdrawn paper. The third is a state of play for every problem, giving what is known, the best current route and the exact open subgoal. The fourth is a record of failure: 207 problems carry a list of the routes tried and the reason each failed. Mathematics has almost no venue for such records, and Tao names "the publication of negative or partial results" among the infrastructure the field now needs [55]. The fifth is the partial results for 142 problems: special cases, correct orders of growth, reductions, and counterexamples to stronger statements. The last is a set of common cores: about 25 problems reduce to three underlying hard problems, so progress on one would move many. These products are unverified in the same sense as the proofs, and the raw working files behind them, about 1300 notes, drafts and scripts, are of little use to anyone. Their value lies in the curated record of each problem. We would release that record per problem, labelled as a sketch, to anyone working on it.

## 14. What the literature changed in our process

We read the closest studies in full and changed our playbooks where they had learned something we had not. The main changes are the following.

*Read the question before answering it.* Copy the source's conventions into the record, check a restatement against known special cases, and treat an easy answer as a sign of misreading: most technically correct answers in the Gemini study were vacuous [15], and AlphaProof's agents proved variants the askers had not meant [25].

*Blind, cross-family referees.* The referee first sees only the question and the proof, as in QED, whose verifier accepted 17 of 214 candidate proofs and had all 17 confirmed by experts [23]; same-family validators overrate their own answers [26]. Review is capped at two rounds, after which the problem goes to the human, since revisions converge on errors the reviewer can no longer see [13].

*A word-by-word statement check*, with a list of the ways a proved statement drifts from the posed one [19, 23], and a citation audit that quotes every cited result.

*Outcome labels and honest reporting*: known, new, fixed statement, or nothing [14]; claimed answers split into flawed, technically correct and meaningfully correct [15]; asymptotic results without explicit thresholds counted as partial [21].

*Neutral briefs*: "prove or disprove", since an agent told to prove a false statement papers over the gap [28]; and repeated narrowing to a smaller question without a new idea counts as no progress [31].

*Human sign-off before release*: the human confirms that the statement matches the question [19], writes or approves an exposition they understand [29], could give a talk on it [55], and adds a disclosure section [50].

## 15. Limits

This is one person's experience in one area, over a few weeks, without a control group. The difficulty and significance scores were assigned by the same agents and by the author, not by independent experts. Correctness has not been audited on a random sample; we intend to do so, and the counts above should be read as counts of claimed answers. The models will be superseded within months. Finally, the area has a dense literature of recent preprints, which made the literature check unusually productive.

## Tool and computational resource disclosure

The work used Claude Code from Anthropic, with Claude Opus 5.5, Sonnet and Haiku, and Codex from OpenAI, with several GPT models; Python with mpmath, SymPy and cvxpy for numerics and certificates, Lean for one formalization, and LaTeX built through a lint gate. This paper was drafted with Claude Code from the project's records and logs and revised by the author. The agent instructions, templates and tools are available in the accompanying process repository.

## Appendix A. A problem record

Each problem is one Markdown file whose YAML header holds the data and whose body holds the memory between sessions. This is the record of a question of Bayart and Yao, abridged; long fields are cut with [...].

```
id: BY-8.7
title: "Minimal commutant for phi = lambda z^p psi (p>=2) on H^2(D):
  can the zero-free assumption on psi in Corollary 8.5 be dropped?"
kind: stated                      # stated | implicit | inference
origin: external                  # external | ours
theme: composition
source:
  paper: "arXiv:2406.19737"
  location: "Question 8.7, p. 33"
  posed_by: [Bayart, Yao]
literature:
  status: open
  checked: 2026-10-13
  evidence: ["arXiv:2406.19737v1 (only version) [...]"]
ours:
  stage: public
  result_scope: full
  sign: proof
  artifacts: [research/composition/solutions/BY-8.7.tex]
  review: "independent agent referee 2026-10-13: correct [...]"
  public: papers/bayart-yao-questions.pdf
  told_authors: null
resolution: {tier: ours-public, ref: "BY-8.7.tex, Theorem 1",
             checked: true, conditional_on: []}
impact: 2
difficulty: 2
scores_note: "[...] difficulty 3->2 after split from BY-8.11 [...]:
  short argument (local Boettcher coordinate [...])"
```

```
tags: [setting:disc, space:hardy, object:composition,
       type:existence, method:construction]
effort: {sessions: 1, last: 2026-10-13, parked: false}
```

The body has four sections: *Statement* (the question with notation defined), *State of play* (rewritten each session), *Failed approaches* (one line per dead route, with its reason) and an *Attempt log* (one line per session: route, outcome, reason, next step).

The fields were filled at different stages and by different hands. At harvest an agent filled the identity and the source: `id`, `title`, `kind`, `origin`, `theme` and `source`. At triage a cheap agent filled `literature` and the scores, `impact`, `difficulty`, `scores_note` and `tags`. During attacks the working agent updated `ours.stage`, `ours.artifacts`, `ours.sign`, the `effort` counters and the body. A referee set `ours.review`, `ours.result_scope` and `resolution`. Only the human set `ours.public` and `ours.told_authors`. Validation by `make problems` rejected unknown fields, invalid values and inconsistent combinations, such as a resolution tier on a partial result.

## Appendix B. How the record fields were used

The database replaced a set of catalogue files and a shared ranking that several workers appended to at once. Two failures forced it: agents repeated routes that had already failed, because the failures lived only in chat transcripts, and two tools with several subagents each needed to edit the list concurrently. Most fields were added one at a time over the following day, each answering a request from the director: a difficulty score "to give us some starting points", "different levels of solved" (the resolution tiers), a way to say which papers bear on a problem, and, a day later, whether we or someone else had asked the question.

To see which fields mattered we counted, over all 772 commits that touched the database, how many changed each field.

| Field | Commits changing it |
|---|---:|
| State of play (prose) | 615 |
| Attempt log (prose) | 566 |
| Stage | 532 |
| Session count and date | about 400 |
| Failed approaches (prose) | 322 |
| Related papers | 319 |
| Artifacts | 304 |
| Literature evidence | 275 |
| Review | 273 |
| Full or partial | 230 |
| Origin, impact, tags | 1–12 |
| Authors told | 3 |
| Claim basis, abandonment reason, aliases | 4 each |

The fields that carried the work were the prose memory between sessions and the few fields that move a problem through the pipeline. Scores, tags and origin were filled once in bulk. Two lessons were negative. The free-text justification of the scores was rewritten in 426 commits, because escalations were appended to it, and so it went stale. The field recording that the askers had been told was changed in only three commits, because nothing in the pipeline required it. Other fields that nothing in the pipeline depended on, such as a free-text relevance line and the minutes spent, were also hardly ever filled in. The lesson is that the prose memory between sessions, together with the handful of fields that move a problem through the pipeline, did the real work; a field that nothing reads or checks is not kept up to date. The field we most missed was how the source poses the question, which we had to reconstruct for Section 5. The schema we now recommend keeps the working fields, freezes

the score justification at triage and logs escalations in the attempt log instead, adds the form of the question, a verbatim quote, the reviewing model and a frozen initial difficulty, makes the publishing step refuse a result whose askers have not been told, and drops the unused fields.

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